

ST AUGUSTINE BUOY 41012, USA

WMO: 996350

Lat: 30.042N

Lon: 80.534W

Elev: 0

StdP: 101.33

Time Zone: -5.00 (W05)

Period: 02-14

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 6.8	(c) 8.8	(d) -1.8	(e) 3.2	(f) 9.9	(g) -0.5	(h) 3.6	(i) 10.3	(j) 16.0	(k) 14.6	(l) 15.0	(m) 13.7	(n) 9.9	(o) 330	(p) 0.518

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 1.9	(c) 29.6	(d) 25.9	(e) 29.2	(f) 25.7	(g) 29.0	(h) 25.6	(i) 27.2	(j) 28.2	(k) 26.7	(l) 28.2	(m) 26.3	(n) 28.2	(o) 4.7	(p) 170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.9	22.6	27.8	26.2	21.6	27.7	25.6	20.9	27.8	85.5	28.2	83.1	28.2	81.4	28.2	28.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 14.2	(b) 12.0	(c) 11.0	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
(a) 14.2	(b) 12.0	(c) 11.0	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	22.6	16.8	17.0	18.9	21.2	23.8	26.3	27.2	27.6	27.1	24.8	21.3	19.0
	DBStd	4.57	4.01	3.26	3.05	2.03	1.30	0.95	0.90	1.16	0.96	2.27	2.76	3.48
	HDD10.0	8	4	2	1	0	0	0	0	0	0	0	0	1
	HDD18.3	213	74	56	31	5	0	0	0	0	0	2	10	36
	CDD10.0	4611	217	197	276	334	429	490	533	545	512	459	338	280
	CDD18.3	1775	28	18	49	90	171	240	275	287	262	203	98	56
	CDH23.3	13572	6	1	4	58	637	2206	2906	3222	2792	1505	204	32
	CDH26.7	2289	0	0	0	0	1	198	560	876	555	98	0	0
Wind	WSAvg	5.9	6.9	6.6	6.2	5.8	5.6	5.0	4.6	4.9	5.7	6.1	7.0	6.8
Precipitation	PrecAvg	1339	102	92	102	77	85	169	160	167	128	102	66	88
	PrecMax	1694	292	265	255	168	230	253	285	321	333	301	141	211
	PrecMin	1013	29	22	15	21	18	58	67	93	38	10	10	31
	PrecStd	190	55	51	57	40	51	54	47	49	62	68	39	49
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.1	23.0	24.0	25.1	26.9	29.1	29.7	30.3	29.4	28.6	26.4	24.8
		MCWB	21.0	20.4	21.7	20.0	23.4	25.5	26.1	26.4	25.2	25.1	22.8	21.8
	2%	DB	23.4	22.2	23.2	24.6	26.3	28.6	29.1	29.8	29.0	28.0	25.8	24.1
		MCWB	20.4	19.4	20.0	20.2	23.3	25.1	25.7	26.1	25.0	24.5	22.1	21.3
	5%	DB	22.7	21.5	22.8	24.1	25.9	28.1	28.9	29.5	28.7	27.7	25.1	23.7
		MCWB	19.7	18.6	19.4	20.3	22.8	24.9	25.6	26.0	24.8	24.2	21.3	20.7
	10%	DB	21.9	20.9	22.3	23.6	25.5	27.8	28.5	29.1	28.5	27.3	24.5	23.1
		MCWB	18.8	17.8	19.0	20.0	22.5	24.7	25.4	25.9	24.6	23.7	20.4	19.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.4	20.9	22.4	22.6	24.7	26.6	27.9	28.0	26.3	25.9	24.1	23.0
		MCDB	23.7	22.3	23.5	23.7	26.1	28.2	28.1	28.5	28.3	27.9	25.8	24.1
	2%	WB	21.2	20.2	20.9	21.9	24.1	26.2	27.1	27.5	26.0	25.4	23.1	22.3
		MCDB	23.0	21.6	22.4	23.4	25.8	27.6	27.7	28.4	28.2	27.5	25.4	23.7
	5%	WB	20.1	19.2	20.4	21.4	23.6	25.8	26.6	26.9	25.7	25.0	22.1	21.6
		MCDB	22.2	20.9	22.1	23.3	25.4	27.4	27.7	28.5	28.0	27.3	24.6	23.2
	10%	WB	19.4	18.1	19.8	20.8	23.1	25.4	26.1	26.5	25.4	24.5	21.3	20.8
		MCDB	21.7	20.2	21.8	23.1	25.0	27.1	27.7	28.6	27.9	27.0	24.2	22.7
Mean Daily Temperature Range	5% DB	MDBR	3.9	3.7	3.0	2.6	1.9	1.8	1.7	1.9	1.8	2.1	2.9	3.7
		MCDBR	3.0	3.2	2.5	2.3	2.0	1.9	1.8	1.8	1.9	2.0	3.0	3.0
		MCWBR	3.5	3.8	2.5	2.7	2.2	1.9	1.6	1.6	1.8	2.0	3.4	3.6
	5% WB	MCDBR	3.2	3.3	2.5	2.6	2.1	1.9	1.6	1.9	1.8	2.1	3.4	3.3
		MCWBR	3.6	4.1	2.8	3.2	2.2	2.1	1.9	2.0	1.8	2.2	4.1	4.0
Clear-Sky Solar Irradiance	taub	0.347	0.358	0.375	0.398	0.434	0.465	0.483	0.471	0.427	0.404	0.365	0.354	
	taud	2.539	2.516	2.473	2.390	2.332	2.314	2.269	2.307	2.419	2.453	2.508	2.540	
	Ebn at Noon	887	907	912	897	861	830	813	821	849	846	858	861	
	Edh at Noon	87	97	107	121	129	130	136	130	112	102	89	84	
All-Sky Solar Radiation	RadAvg	3.10	3.81	5.05	6.27	6.73	6.57	6.58	6.05	5.05	4.34	3.38	2.82	
	RadStd	0.23	0.38	0.40	0.31	0.43	0.58	0.34	0.36	0.43	0.34	0.23	0.23	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (26 neighbors)		+0.44	+0.76	N/A	N/A	N/A	N/A	-9	-62	+160	+116

Nomenclature: See separate page